

CYBER SECURITY COURSE

Independent Windows Incident Investigation

Professional Case Report 20

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Case ID	CSC-2026-20
Discipline	Incident response
Overall risk	Critical
Evidence status	Confirmed by saved Redline evidence

Purpose

Demonstrate independent scoping, evidence correlation and reporting when the initial alert provides limited context.

Scope note

This report documents work performed in an authorized training environment. Confirmed findings are separated from methodology and expected laboratory behavior.

Executive Summary

An investigation began without a supplied indicator list. The analyst had to connect suspicious files, persistence data and PowerShell activity into one incident narrative. The independent workflow produced a stronger incident narrative than a checklist alone. Each pivot was based on an observed artifact and extended the scope in a controlled way.

Case Profile

Scenario	An investigation began without a supplied indicator list. The analyst had to connect suspicious files, persistence data and PowerShell activity into one incident narrative.
Environment	Windows Redline evidence set covering file system, Registry, persistence and Event Logs.
Tools	Mandiant Redline, Timeline analysis, Registry forensics, PowerShell event analysis, IOC development

Methodology

I used the following sequence to keep the analysis repeatable and to separate observation from interpretation.

1. Started with time and path anomalies rather than a predetermined answer.
2. Pivoted from malicious file paths into Registry persistence.
3. Correlated common modification times across autorun evidence.
4. Reviewed PowerShell events within the confirmed compromise window.
5. Separated facts, interpretation and recommended response actions.

Technical Findings

ID	Finding	Risk	Evidence and analysis
F01	File-to-persistence correlation	Critical	The same attacker-controlled path appeared in both file-system and Run-key evidence.
F02	Timeline convergence	High	Related artifacts clustered between 02:53:47 and 03:17:55 UTC on 1 June 2017.
F03	PowerShell follow-on activity	High	PowerShell records showed archive extraction and additional comm within the compromise window.
F04	Eight malicious or dual-use artifacts	Critical	The evidence set documented a broader toolset than the first suspicious binary alone.

Representative Commands and Actions

The following commands or analysis actions represent the core of the practical workflow.

Pivot by path, hash, timestamp and Registry text value

Maintain an investigation timeline with UTC timestamps

Risk Assessment

Risk	Level	Operational impact
Incomplete scoping	Critical	Focusing on one file leaves persistence and tools undiscovered
Repeated compromise	High	Unremoved autorun mechanisms restore attacker access

Recommendations

1. Treat related paths, hashes, accounts and timestamps as an IOC set.
2. Search peer systems for the same evidence before declaring containment.
3. Preserve a UTC timeline with source references.
4. Rebuild the endpoint when full attacker scope cannot be proven.

Knowledge Demonstrated

- Hypothesis-driven investigation
- Cross-artifact correlation
- Incident timeline
- Executive reporting

Conclusion

The independent workflow produced a stronger incident narrative than a checklist alone. Each pivot was based on an observed artifact and extended the scope in a controlled way.

Hands On Lab Practical 20

Completed as part of the authorized cyber security course practical work.